

Simple Dictionary – Java Mini Project

Definition

A Simple Dictionary is a beginner-friendly Java mini project that stores words and their meanings. The user can enter a word, and the program searches the dictionary and displays its meaning.

Main Purpose

The main purpose of this project is to store and search word-meaning pairs efficiently.

Example

Word: Computer

Meaning: An electronic device used to process data.

Main Features

1. Add a word – Add a new word and its meaning.
2. Search a word – Find the meaning of a given word.
3. Display all words – Show all stored words and meanings.
4. Exit – Close the program.

Java Concepts Used

- Classes and Objects
- Methods
- Scanner – for user input
- HashMap – to store words and meanings
- Loops
- if-else / switch
- String handling

Simple Java Code

```
import java.util.HashMap;
import java.util.Scanner;

public class SimpleDictionary {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        HashMap<String, String> dictionary = new HashMap<>();

        dictionary.put("computer", "An electronic device used to process data.");
        dictionary.put("java", "A programming language.");
        dictionary.put("array", "A collection of elements of the same data type.");

        while (true) {

            System.out.println("\n--- Simple Dictionary ---");
            System.out.println("1. Search Word");
            System.out.println("2. Add Word");
            System.out.println("3. Display All Words");
            System.out.println("4. Exit");

            System.out.print("Enter your choice: ");
            int choice = sc.nextInt();
            sc.nextLine();
        }
    }
}
```

```

    if (choice == 1) {

        System.out.print("Enter word: ");
        String word = sc.nextLine().toLowerCase();

        if (dictionary.containsKey(word)) {
            System.out.println("Meaning: " + dictionary.get(word));
        } else {
            System.out.println("Word not found.");
        }

    } else if (choice == 2) {

        System.out.print("Enter word: ");
        String word = sc.nextLine().toLowerCase();

        System.out.print("Enter meaning: ");
        String meaning = sc.nextLine();

        dictionary.put(word, meaning);

        System.out.println("Word added successfully.");

    } else if (choice == 3) {

        System.out.println("\nWords and Meanings:");

        for (String word : dictionary.keySet()) {
            System.out.println(word + " : " + dictionary.get(word));
        }

    } else if (choice == 4) {

        System.out.println("Thank you!");
        break;

    } else {
        System.out.println("Invalid choice.");
    }
}

sc.close();
}
}

```

How the Project Works

1. Create Dictionary

```
HashMap dictionary = new HashMap<>();
```

HashMap stores data in key-value pairs:

Key → Value

java → A programming language

computer → An electronic device

array → Collection of elements

2. Add Words

```
dictionary.put("java", "A programming language.");
```

The put() method adds a word and its meaning.

3. Search

```
dictionary.containsKey(word)
```

Checks whether the word exists.

```
dictionary.get(word)
```

Gets the meaning of the word.

4. Display

```
for (String word : dictionary.keySet())
```

This goes through all the words stored in the dictionary.

Example Output

```
--- Simple Dictionary ---
1. Search Word
2. Add Word
3. Display All Words
4. Exit

Enter your choice: 1
Enter word: java

Meaning: A programming language.
```

Why This Is a Good Mini Project

It is simple enough for a beginner, but it demonstrates an important DSA concept: HashMap, where searching for a word is generally very fast.

Presentation Order

Introduction → Objective → Features → Technologies/Concepts → Algorithm/Working → Java Code → Output → Advantages → Future Enhancements → Conclusion.